HIRE AI : A Rule-Based Intelligent Recruitment Expert System

A PROJECT REPORT

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CHENNAI

BONAFIDE CERTIFICATE

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ABSTRACT

HiringAI – AI-Based Recruitment System is a web-based application that simplifies recruitment by matching candidates' resumes with job requirements using classical Artificial Intelligence techniques. It automates resume screening and candidate ranking, reducing the time and effort required in traditional recruitment.

The system provides separate interfaces for candidates and HR personnel. Candidates can upload resumes, while HR personnel can define job requirements and identify suitable candidates based on skills, qualifications, and experience. It uses rule-based expert systems, knowledge representation, reasoning, and constraint satisfaction to rank candidates according to predefined criteria.

The application uses React with Vite for the frontend, Fast API for the backend, and Supa base with PostgreSQL for database management. Unlike machine learning and deep learning approaches, it uses transparent, predefined rules to ensure explainable candidate evaluations.

The main objective of HiringAI is to reduce manual screening, improve recruitment efficiency, and support faster, consistent, and organized hiring decisions.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The modern corporate recruitment landscape has undergone extensive digital transformation over the past decade. With standard job postings frequently attracting hundreds to thousands of candidate applications across multinational enterprises, traditional manual screening of resumes has become prohibitively time-consuming, labor-intensive, and prone to recruiter fatigue. To accelerate preliminary talent screening, modern enterprises increasingly rely on automated Applicant Tracking Systems (ATS).

However, prevailing commercial ATS solutions rely heavily on probabilistic machine learning, statistical vector embeddings, and generative Large Language Models (LLMs). While scalable, these probabilistic approaches introduce severe vulnerabilities into enterprise hiring pipelines: they operate as uninterpretable 'black-boxes', exhibit non-deterministic scoring fluctuations across identical document submissions, hallucinate non-existent candidate qualifications, and systematically perpetuate historical demographic biases.

Furthermore, stringent international legal frameworks—including the European Union Artificial Intelligence Act (EU AI Act, Regulation 2024/1689), which classifies AI systems used for recruitment and candidate evaluation as High-Risk AI Systems—now strictly mandate complete explainability, transparency, mathematical auditability, and verifiable non-discrimination. The HireAI platform addresses these critical industry and regulatory imperatives by establishing a First-Order Deterministic Explainable AI (FOAI) expert architecture that eliminates stochastic scoring in favor of verifiable, rule-based inference.

1.2 PROBLEM STATEMENT

Traditional automated talent evaluation platforms face several technical and operational challenges that affect recruitment transparency, fairness, security, and efficiency. These limitations can reduce candidate trust, complicate recruiter decision-making, and increase the risk of inaccurate or inconsistent hiring outcomes. The major deficiencies identified in existing recruitment systems are described below.

- **Black-Box Opacity and Non-Determinism:** Many AI-based screening systems generate candidate suitability scores without providing sufficient explanations of the factors influencing their results. The absence of transparent evaluation criteria and traceable scoring mechanisms makes it difficult for recruiters to verify recommendations and for candidates to understand how their qualifications were assessed.

- **Algorithmic Bias and Hallucination:** AI models trained on historical recruitment data may reproduce existing biases related to gender, background, educational institutions, or previous hiring practices

- **Adversarial Manipulation and Prompt Injection:** AI-assisted resume screening systems may be vulnerable to manipulated documents containing hidden instructions, excessive keyword repetition, or embedded prompt-injection attempts. Such content can influence poorly designed screening workflows and produce misleading evaluation results if uploaded documents are not properly validated and processed as untrusted input.

- **Persistent Storage and Personal Information Exposure:** Recruitment platforms process sensitive information, including contact details, employment history, educational qualifications, and resumes. Inadequate storage controls, excessive data retention, or improperly configured storage services can expose this information to unauthorized access.

These challenges highlight the need for a recruitment platform that combines explainable candidate evaluation, bias-aware screening, secure resume processing, privacy protection, and centralized application tracking. HireAI aims to address these limitations through structured resume analysis, transparent scoring criteria, controlled access to candidate data, and an integrated recruitment workflow

1.3 OBJECTIVES OF THE PROJECT

The primary objective of HireAI is to develop an intelligent, secure, and transparent recruitment platform that simplifies resume screening, improves candidate evaluation, and streamlines recruitment workflow management. The specific objectives of the project are outlined below.

- **Deterministic FOAI Ranker Engine:** To develop and evaluate a First-Order Artificial Intelligence (FOAI) expert system that calculates candidate match percentages using predefined rules and mathematical scoring criteria. The engine aims to provide consistent, reproducible evaluation results without relying on third-party AI API dependencies.
- **Zero-Disk Volatile Memory Processing:** To process candidate resumes in volatile memory buffers using mechanisms such as io.BytesIO, minimizing temporary disk storage and reducing the risk of sensitive documents remaining on server storage after processing. This approach supports privacy-focused resume handling.
- **Five-Pillar Explainable ATS Health Audit:** To evaluate resume quality across five essential dimensions: Format and Parsability, Technical Keyword Relevance, Measurable Impact, Section Completeness, and Contact Hygiene. The audit provides structured feedback to help candidates identify weaknesses and improve the presentation of their qualifications.
- **Bi-Directional Identity Verification:** To implement candidate identity and resume consistency checks that help identify mismatches between submitted profile information and resume details. The objective is to reduce identity misrepresentation, detect inconsistencies, and discourage unauthorized or repetitive application submissions.
- **Enterprise Kanban Pipeline Synchronization:** To establish a centralized recruitment workspace with role-based access, interactive candidate leaderboards, inspectable scoring traces, and structured Kanban stage transitions. This enables recruiters to monitor application progress, coordinate hiring activities, and maintain consistent recruitment records.

Through these objectives, HireAI seeks to deliver a reliable, explainable, and privacy-conscious recruitment solution that combines rule-based evaluation, secure document processing, and centralized hiring management.

1.4 SCOPE AND BOUNDARIES

The scope of Phase I encompasses the complete end-to-end recruitment lifecycle within enterprise environments:

- **Candidate Portal:** Provides live view-only job browsing, compulsory multi-field profile completion, in-memory resume parsing, real-time 5-pillar ATS health auditing, interactive PDF preview, and rule-gated application submission.
- **HR / Enterprise Recruiter Portal:** Enables real-time job requisition creation, deterministic candidate match ranking, drill-down explainable proof trace inspection, and interactive Kanban pipeline status management.
- **Backend FOAI Microservice:** High-performance Fast API REST API, Supa base Cloud Storage abstraction, deterministic AST parser, multi-domain skill taxonomy, and stateless JWT token authentication.

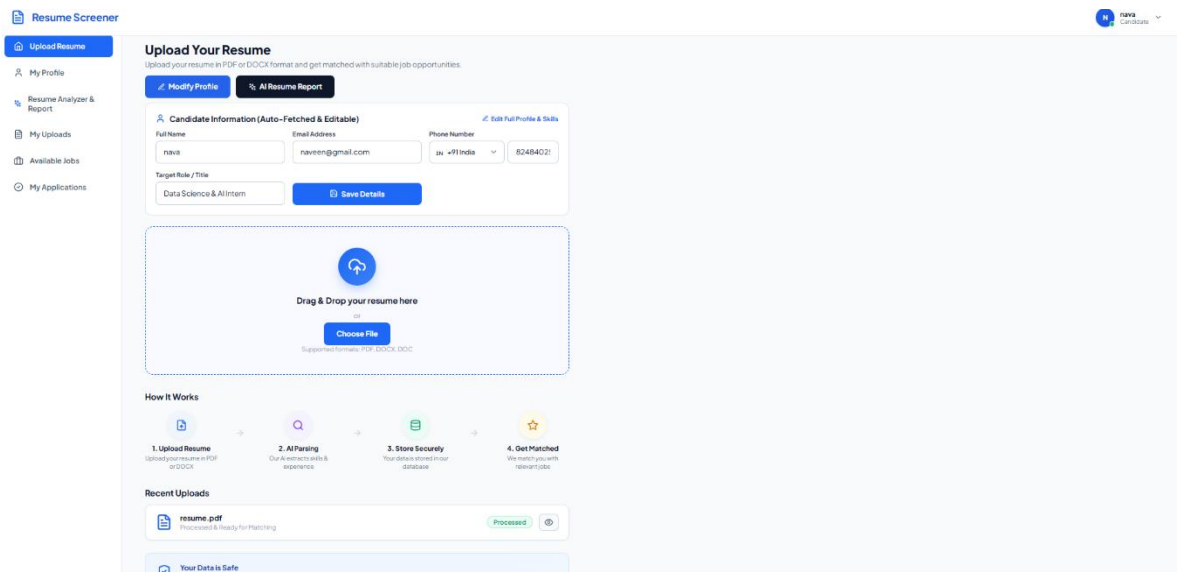


Figure 1.1 Candidate Portal of the HireAI System

Figure 1.1 shows the Upload Resume page of the Candidate Portal. It combines an editable candidate information panel (name, email, phone number and target role), a drag-and-drop area accepting PDF, DOCX and DOC files, a four-step “How It Works” summary, and a Recent Uploads list in which the file cyber_resume.pdf is marked as “Processed”. The page represents the entry point of the candidate workflow described in this section.

The recruiter-facing counterpart of the Candidate Portal is the HR Portal, whose central component is a Kanban-style hiring board. Figure 1.2 presents this board, which consolidates the applications received for a requisition into stage-wise columns.

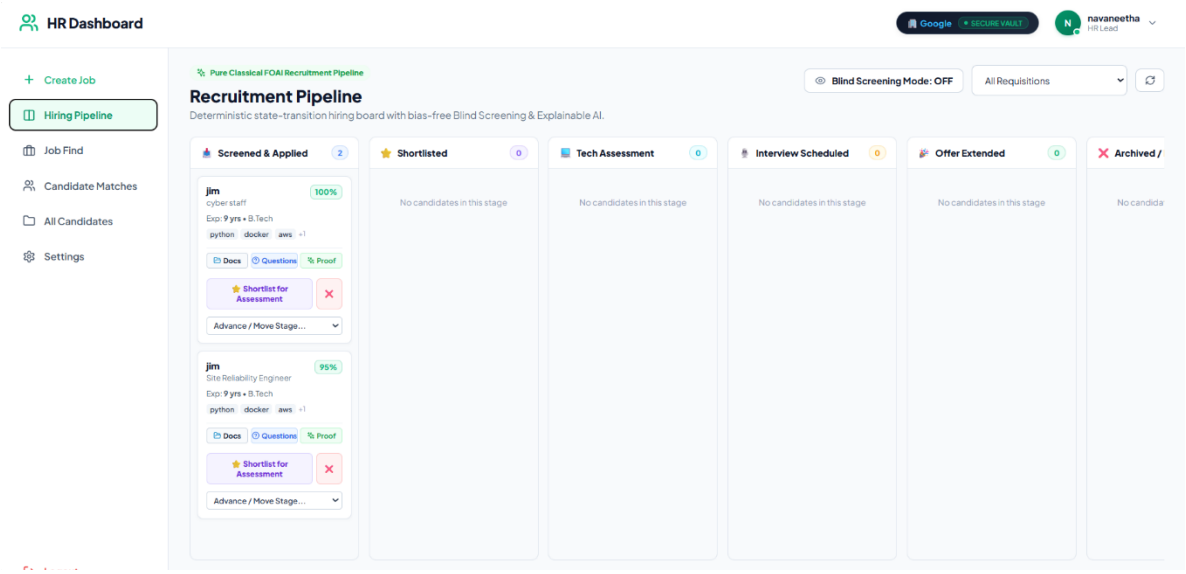


Figure 1.2 HR Recruitment Pipeline Dashboard of the HireAI System

Figure 1.2 shows the Recruitment Pipeline of the HR Portal, in which applications are arranged in six stage columns: Screened & Applied, Shortlisted, Tech Assessment, Interview Scheduled, Offer Extended and Archived / Rejected. Each column displays the number of candidates it contains. Every candidate card presents the applied role, match percentage, experience, qualification and extracted skills, along with shortcuts to documents, screening questions and the proof view. Cards also carry stage-specific action buttons, a rejection control and an “Advance / Move Stage” selector. The header provides a requisition filter, a refresh control and a Blind Screening Mode toggle, which is shown in the OFF state in this capture. The displayed candidates are sample records prepared for demonstration, so the figure illustrates the structure and workflow of the recruiter interface and not the outcome of a live hiring process.

1.5 SECURITY REQUIREMENTS & CONSIDERATIONS

In accordance with cybersecurity engineering best practices, HireAI incorporates multiple defensive safeguards to protect sensitive candidate information, maintain application integrity, and ensure secure access to recruitment functionalities.

• Confidentiality: Zero local disk caching is maintained during resume processing. Resumes are handled using in-memory byte buffers and transmitted to Supa base encrypted storage. Access to candidate information is restricted to authorized users to minimize the risk of data exposure.

• Integrity: Deterministic Abstract Syntax Tree (AST) parsing processes supported document structures without executing embedded scripts. Input validation and structured parsing help reduce risks associated with malformed documents and unsafe file processing.

• Availability & DoS Protection: Strict MIME-type validation and a maximum payload size of 10 MB help prevent unsupported uploads and excessive resource consumption. Request validation and rate limiting further help protect application endpoints against denial-of-service (DoS) risks.

• Authentication & Authorization: Stateless JSON Web Tokens (JWTs) provide authenticated access, while Role-Based Access Control (RBAC) separates candidate operations from recruiter and administrative privileges. Authorization checks help prevent unauthorized access to restricted resources.

• Secure File Uploads: Uploaded resumes are validated for permitted formats and size restrictions before processing. Invalid or malformed files are rejected to reduce the risk of file-upload vulnerabilities and parsing errors.

• Database & API Security: Parameterized database queries and validated API requests help prevent SQL injection and unauthorized operations. Access permissions follow the principle of least privilege, restricting each component to the resources required for its functions.

These security measures help strengthen HireAI against common cybersecurity threats while supporting the confidentiality, integrity, and availability of candidate data and recruitment services.

1.6 SECURITY REQUIREMENTS AND CONSIDERATIONS

HireAI incorporates essential cybersecurity measures to ensure the confidentiality, integrity, and availability of sensitive candidate information. Since the platform handles personal details and resumes, appropriate safeguards are necessary to prevent unauthorized access, data leakage, and potential security threats.

• Confidentiality: Resume files are processed using in-memory buffers to minimize local data storage, while Supa base provides storage for candidate records. Access to sensitive information is restricted according to user roles to maintain privacy.

• Integrity: Deterministic Abstract Syntax Tree (AST) parsing processes supported document structures without executing embedded scripts. Input validation helps identify malformed files and reduces risks associated with unsafe document processing.

• Availability and DoS Protection: Strict MIME-type validation and a 10 MB file size limit help prevent unsupported uploads and excessive resource consumption. Request validation and rate limiting can further reduce the risk of Denial-of-Service (DoS) attacks.

• Authentication and Authorization: JSON Web Token (JWT)-based authentication and Role-Based Access Control (RBAC) separate candidate operations from recruiter and administrative privileges. Authorization checks help ensure that users can access only permitted features and resources.

• Database and API Security: Parameterized queries help prevent SQL injection, while server-side validation ensures that incoming requests meet expected requirements. Secure API endpoints reduce the possibility of unauthorized operations.

• Data Protection: HTTPS/TLS protects information during transmission, while appropriate storage encryption and access controls help safeguard candidate records. Sensitive credentials and API keys should be stored securely and excluded from publicly accessible source code.

• Monitoring and Error Handling: Controlled error messages prevent unnecessary disclosure of internal system details. Security logs can help identify suspicious requests, repeated authentication failures, and other unusual activities.

These security requirements collectively strengthen HireAI's defensive capabilities, protect candidate privacy, reduce common application vulnerabilities, and support the reliable operation of the recruitment platform.

1.7 DATA PRIVACY AND SECURE STORAGE

HireAI prioritizes data privacy by protecting sensitive candidate information throughout the recruitment process. Personal details, resumes, and recruitment records are handled securely and accessed only by authorized users. Supabase is used for data storage, with appropriate access policies to prevent unauthorized retrieval, modification, or disclosure of records. Secure communication protocols such as HTTPS/TLS help protect information transmitted between the client application and backend services.

- **Access Control:** Role-Based Access Control (RBAC) restricts access according to user responsibilities. Candidates can access their own information, while recruiters are permitted to view candidate records only within their authorized scope.
- **Secure Data Transmission:** HTTPS/TLS helps protect sensitive information exchanged between the frontend, backend, and supported external services, reducing the risk of interception during transmission.
- **Storage Protection:** Appropriate database permissions and storage access policies help safeguard resumes and personal records. Where supported, encryption at rest provides an additional layer of protection against unauthorized data exposure.
- **Data Minimization:** Only information necessary for recruitment-related operations should be collected and processed. Limiting unnecessary data collection reduces privacy risks and helps prevent excessive exposure of personal information.
- **Privacy and Retention:** Candidate information should be retained only for the required period and deleted when it is no longer needed, subject to applicable requirements. Secure deletion procedures help reduce the risk of unnecessary long-term data exposure.
- **Secure Configuration:** Database credentials, API keys, and other sensitive configuration values should be protected from public access. Regular reviews of storage permissions and access policies help identify potential security weaknesses.

Through these measures, HireAI aims to maintain candidate confidentiality, prevent unauthorized access, and support responsible data handling throughout the recruitment lifecycle. Effective privacy controls and secure storage practices strengthen user trust and contribute to the reliability of the platform.

1.8 SDG ALIGNMENT

HireAI aligns with the United Nations Sustainable Development Goals (UN SDGs) 8, 9, 10, and 16 by promoting fair recruitment, equal employment opportunities, technological innovation, and transparent hiring practices. The platform integrates artificial intelligence into recruitment workflows to simplify resume screening, improve candidate evaluation, and support informed hiring decisions. By combining technological advancement with responsible data management, HireAI aims to contribute to a more efficient, inclusive, and trustworthy recruitment ecosystem.

- **SDG 8 – Decent Work and Economic Growth:** HireAI supports SDG 8 by streamlining recruitment activities and helping recruiters identify candidates whose skills, qualifications, and experience match job requirements. Automated resume screening reduces repetitive manual work and enables recruiters to process applications more efficiently. By simplifying the initial screening process, the platform can help connect suitable candidates with relevant employment opportunities and support a more productive recruitment environment.
- **SDG 10 – Reduced Inequalities:** HireAI aims to promote inclusive recruitment by emphasizing job-related qualifications, skills, and experience during candidate evaluation. Structured screening criteria can help reduce reliance on subjective initial assessments and encourage consistent consideration of applicants. However, AI systems must be monitored for potential bias to ensure that automated recommendations do not unfairly disadvantage candidates from different backgrounds.

Furthermore, HireAI emphasizes responsible AI adoption by recognizing the importance of human supervision, data privacy, fairness, and accountability in automated recruitment. AI-generated results should assist recruiters in decision-making rather than independently determine employment outcomes. Regular evaluation of screening criteria can help identify potential limitations and improve the reliability of the system.

Overall, HireAI contributes to the broader objectives of sustainable employment, digital innovation, social inclusion, and responsible governance. By combining recruitment automation with appropriate security controls and ethical evaluation practices, the platform aims to support a more accessible, efficient, and equitable hiring process.

CHAPTER 2
LITERATURE REVIEW

2.1 EXISTING SYSTEMS WITH SECURITY APPROACHES

Modern talent recruitment software has evolved through three historical generations, each characterized by specific algorithmic trade-offs:

1. First-Generation Keyword ATS Systems (e.g., Taleo, Jobvite): Early automated recruitment tools relied on literal substring matching and SQL queries to search for candidate resumes containing exact keyword strings. While computationally deterministic, these systems are easily fooled by keyword stuffing, fail to capture contextual synonyms, and lack structured reasoning.

2. Second-Generation Statistical NLP & Embedding Models (e.g., Word2Vec, BERT): Subsequent systems incorporated dense vector semantic representations (Word2Vec, GloVe, BERT, SBERT) to compute cosine similarity between job descriptions and resume text. While semantic understanding improved, these systems introduced severe opacity, high computational overhead, and non-linear scoring behaviors.

3. Third-Generation LLM & Generative AI Screeners (e.g., GPT-4 Powered ATS): Recent commercial solutions employ generative Large Language Models (GPT-4, Claude, LLaMA) to evaluate candidates via natural language prompts. However, LLMs suffer from severe security vulnerabilities: non-deterministic temperature drift, prompt injection attacks, resume text hallucinations, and astronomical operational API costs.

2.2 RESEARCH GAPS & VULNERABILITIES

A comprehensive review of literature reveals four fundamental research gaps in automated hiring:

- **Lack of Explainability & Proof-Traces:** Existing platforms output an opaque percentage score (e.g., '78% match') without verifiable mathematical reasoning, leaving recruiters unable to defend decisions during labor compliance audits.
- **Vulnerability to Document Identity Mismatch:** Current screeners evaluate resume content in isolation from applicant account details, enabling candidates to submit third-party resumes under forged profile names.
- **Absence of Pre-Application Gap Simulation:** Candidates have no means of understanding skill gaps before applying, leading to massive volumes of unqualified applications.
- **Non-Compliance with High-Risk AI Legislation:** Black-box AI models fail the strict audibility and anti-discrimination standards established by the EU AI Act and EEOC guidance.

Table 2.1 Comparative Analysis of Recruitment Systems & Algorithmic Models

System Metric / Feature	Keyword ATS	Vector / BERT ATS	LLM Screeners	HireAI (Proposed FOAI)
Scoring Determinism	100% Deterministic	Stochastic (Thresholds)	Non-Deterministic	100% Deterministic & Verifiable
Explainable Proof Traces	Partial (Matched Keywords)	None (Opaque Latent Space)	Natural Language (Hallucinated)	Complete Mathematical Proofs
Prompt Injection Defense	High (No LLM Engine)	High (Embedding Only)	Extremely Vulnerable	Immune (Deterministic AST)
Zero-Disk RAM Security	No (Disk Persistence)	No (Disk Persistence)	No (Third-Party Cloud)	Yes (io.BytesIO RAM Buffers)
Inference Latency	50 - 150 ms	300 - 1,200 ms	1,500 - 5,000 ms	< 45 ms (Sub-50ms Ultra-Fast)

2.3 PROPOSED HIREAI SYSTEM

2.3.1 Overview of the Proposed System

HireAI is engineered as a zero-disk, first-order deterministic talent acquisition and audit platform. It bridges the gap between automated recruitment efficiency and verifiable legal compliance by executing classical AI production rules, multi-domain skill taxonomies, and strict identity cross-auditing across dual React portals and a Fast API microservice.

2.3.2 Objectives of the Proposed System

The proposed architecture achieves complete algorithmic transparency by decomposing candidate matching into four explicit, rule-governed dimensions: Technical Skill Coverage (50%), Relevant Experience Alignment (25%), Formal Education Attainment (15%), and Domain Certification Credentials (10%). Every evaluation produces a comprehensive, inspectable proof trace detailing matched requirements, missing competencies, and mathematical score derivations.

2.3.3 Key Advantages of Proposed Architecture

- **1. Pure Classical FOAI:** Requires zero external third-party AI API calls, eliminating ongoing operational API expenditures and preventing external data leakage.
- **2. Sub-50ms Processing Latency:** Executes full document extraction, AST line parsing, and multi-pillar rule evaluation in under 50 milliseconds per candidate.
- **3. Bi-Directional Identity Auditing:** Detects and penalizes candidate-to-resume identity mismatches, preventing credential fraud and account spoofing.

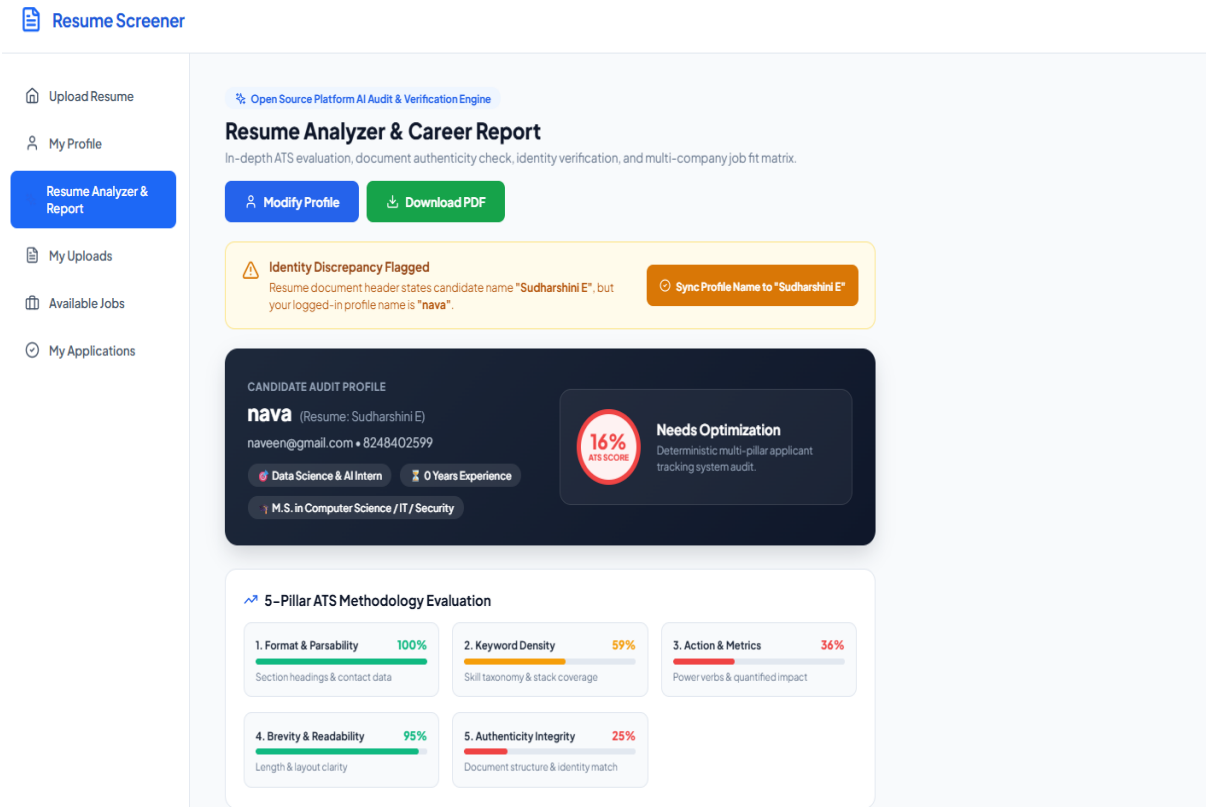


Figure 2.1 Proposed Explainable Resume Evaluation Interface

Figure 2.1 shows the Resume Analyzer & Career Report page, which presents the output of the five-pillar ATS health audit. The page displays a candidate audit profile, an overall ATS score, and the individual pillar scores for Format & Parsability (95%), Keyword Density (57%), Action & Metrics (53%), Brevity & Readability (95%) and Authenticity Integrity (10%). In this capture, the system raises an “Invalid / Non-Resume Document Detected” warning and reports an overall score of 10%, because the uploaded file lacked standard career sections. The figure therefore demonstrates the document-validation behavior and the per-pillar score presentation for a non-conforming document, and it does not represent the evaluation of a well-formed resume.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The HireAI platform employs a decoupled, microservice-oriented client-server architecture designed for high throughput, absolute mathematical determinism, and end-to-end cybersecurity. The architecture is logically partitioned into five discrete operational tiers: Input Ingestion Tier, Preprocessing & AST Sanitization Tier, Expert System Core, Evaluation & Proof Trace Engine, and Dual Output Portals.

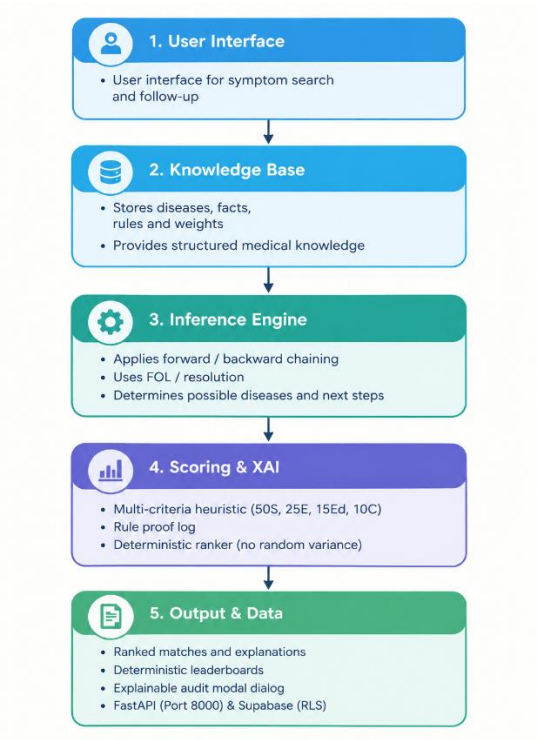


Figure 3.1 HireAI System Architecture & Multi-Tier Processing Pipeline

The comprehensive operational workflow across these tiers comprises the following structured layers:

- **Client Presentation Layer:** Dual responsive single-page web applications built with React 18, Vite, and Tailwind CSS. The Candidate Portal (Port 5173) delivers profile management, ATS health auditing, interactive PDF preview, and rule-gated application submission. The HR Recruiter Portal (Port 5174) delivers job requisition management, candidate match ranking, drill-down proof traces, and Kanban pipeline stage orchestration.

• **API Gateway & Security Tier:** Asynchronous Fast API REST microservice enforcing stateless JWT validation, Role-Based Access Control (RBAC), and CORS security policies. The API gateway receives multipart/form-data byte streams and routes them directly to volatile processing pipelines.

• **Deterministic FOAI Core Engine:** Contains the In-Memory Resume Parser, Technical Skill Taxonomizer, 5-Pillar ATS Health Audit Engine, and Forward-Chaining Inference Ranker. Operates with 100% determinism on in-memory buffers.

3.1.1 Hardware and Software Specifications

Table 3.1 Hardware and Infrastructure Specifications

Hardware Component	Minimum Requirement	Recommended Specification
Processor (CPU)	Dual-Core 2.0 GHz (x86_64 / ARM64)	Intel Core i5 / i7 or AMD Ryzen 5 / Apple M-series
System Memory (RAM)	4 GB DDR4	8 GB / 16 GB DDR4/DDR5 High-Speed RAM
Storage Space	500 MB Available Storage	2 GB NVMe SSD Storage
Network Interface	Standard Broadband (10 Mbps)	High-Speed Broadband / Gigabit Ethernet

Table 3.2 Software Environment and Technology Stack

Software Layer / Component	Technology / Framework	Role / Operational Purpose
Frontend UI Framework	React 18.2, Vite 5, TailwindCSS	Dual responsive SPAs for Candidates & HR Recruiters
Backend REST Microservice	FastAPI (Python 3.10+), Uvicorn	Asynchronous API routing, JWT auth, RBAC enforcement
PDF & DOCX Parsers	pypdf 4.0, python-docx 1.1	In-memory binary stream extraction and text sanitization
Security & Cryptography	PyJWT, BCrypt, Passlib	Stateless authentication token signing, password hashing
Cloud Persistence	Supabase (PostgreSQL & Storage Vault)	Secure remote resume storage and structured relational data

3.2 MODULE DESIGN

HireAI is architected into five decoupled core operational modules:

- **Auth & Security Module:** Handles candidate and recruiter registration, BCrypt password hashing, stateless JWT issuance, and RBAC route guarding.
- **In-Memory Resume Ingestion & Parsing Module:** Accepts binary PDF/DOCX byte streams, extracts text via PyPDF2/python-docx into RAM buffers, and sanitizes input strings.
- **FOAI Candidate Ranker & Skill Taxonomizer:** Categorizes extracted skills into 40+ multi-domain taxonomies (Frontend, Backend, Cloud, Cybersecurity, AI/ML, DevOps) and calculates weighted match scores.
- **Multi-Pillar ATS Health Audit Module:** Evaluates format parsability, keyword density, measurable metrics, section completeness, and contact integrity.
- **Cross-Portal Kanban Pipeline Synchronization Module:** Enforces strict stage progression rules across APPLIED, SHORTLISTED, TECHNICAL_INTERVIEW, HR_INTERVIEW, and OFFERED states.

Figure 3.2 shows the candidate-side view of the application lifecycle, while the recruiter-side board is shown in Figure 1.2.

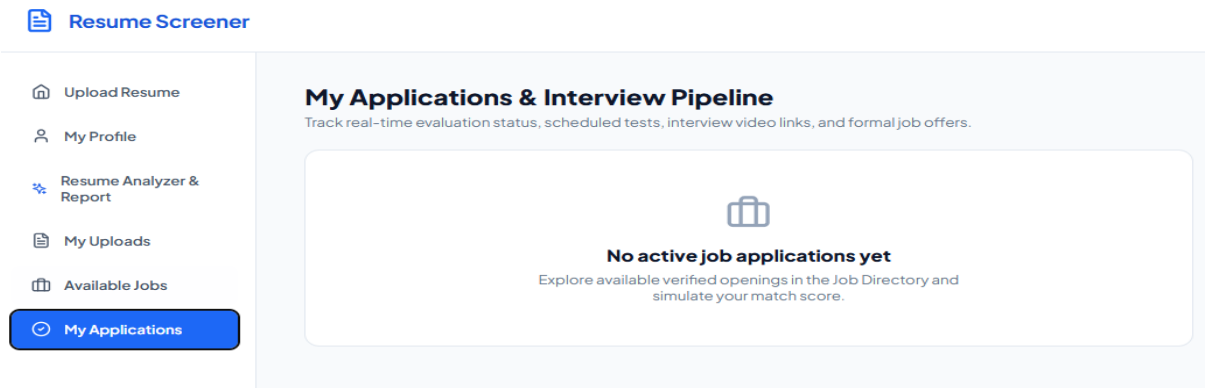


Figure 3.2 Candidate Application Tracking and Interview Pipeline Interface

Figure 3.2 shows the “My Applications & Interview Pipeline” page of the Candidate Portal. Each application is displayed with its organization, location, application date and current status badge, together with a five-step progress tracker (Applied, Shortlisted, Tech Test, Interview and Job Offer) that broadly corresponds to the lifecycle states enforced by the module

3.3 DATA FLOW & UML DIAGRAMS

The system data flow is formalized through standard structural and behavioral architectural models:

- **Level 0 Context DFD:** Illustrates the primary boundary where Candidate and HR users interact with the HireAi Gateway via secure HTTPS/REST calls.
- **Level 1 Modular DFD:** Details the interaction between authentication services, in-memory stream parsers, rule evaluation engines, and persistence layers.
- **Use Case Diagram:** Formalizes actor permissions: Candidates manage profiles, upload resumes, view ATS scores, and submit applications; HR recruiters create job requisitions, inspect rank traces, and advance candidate stages.

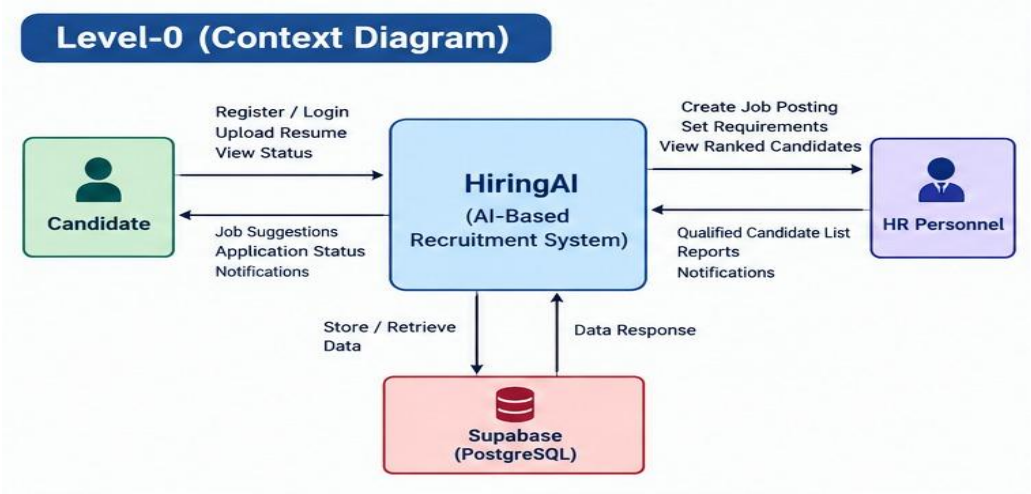


Figure 3.3 Level-0 DFD: Illustrates the overall data flow between the Candidate, HR Personnel, HireAI recruitment system, and Supabase (PostgreSQL) database

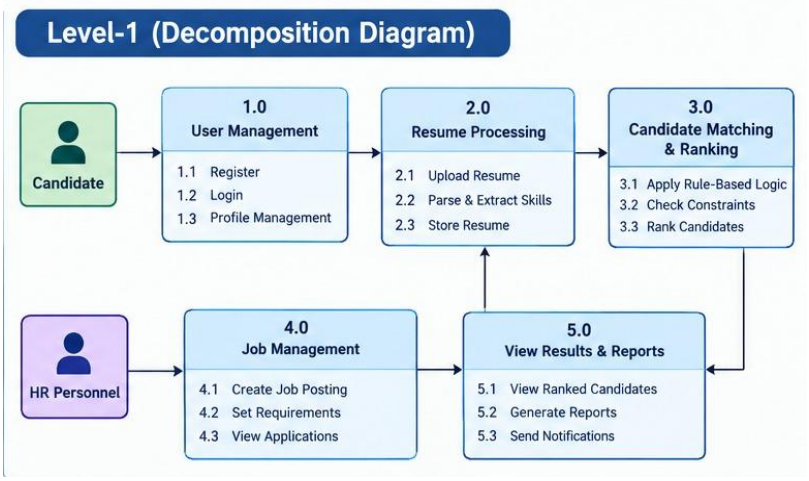


Figure 3.4 Level-1 DFD: Represents the detailed decomposition of the HireAI system into user management, resume processing, candidate matching and ranking, job management, and results and reports.

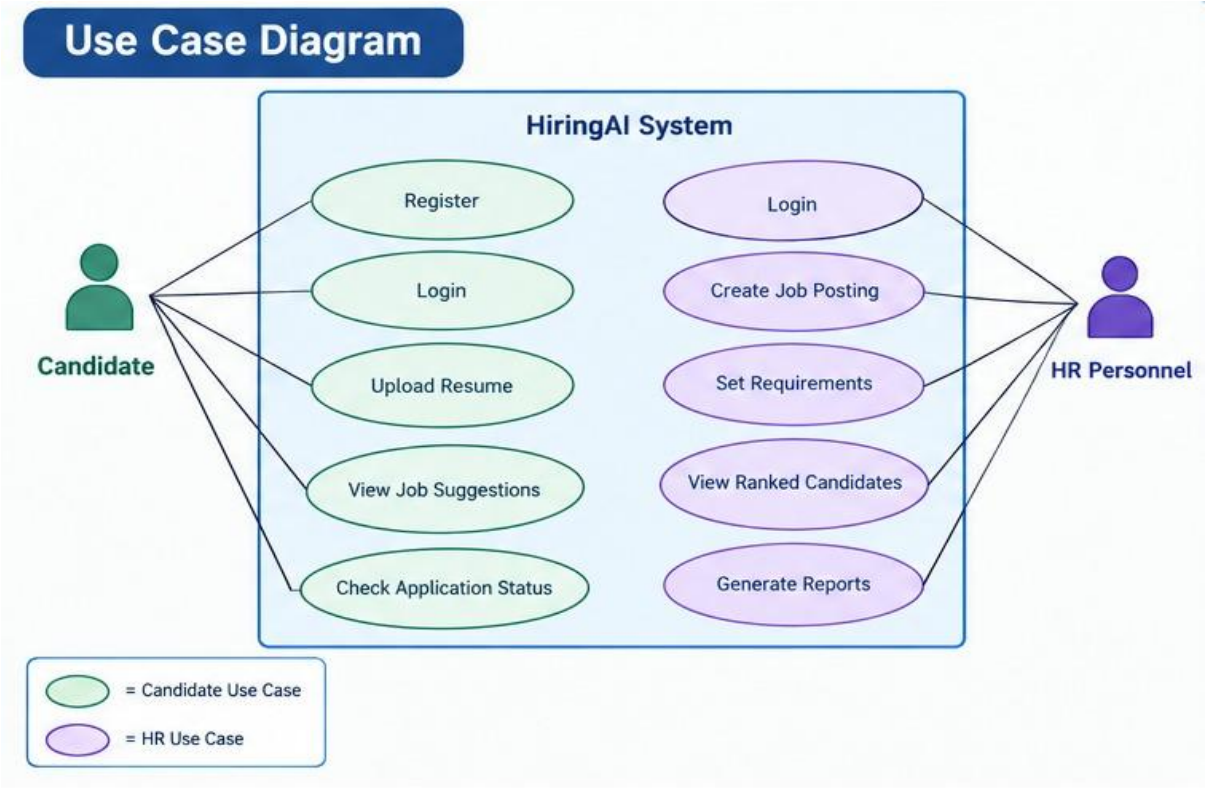


Figure 3.5 Use Case Diagram: Depicts the interactions of Candidate and HR Personnel with the HireAI system and the functionalities available to each actor.

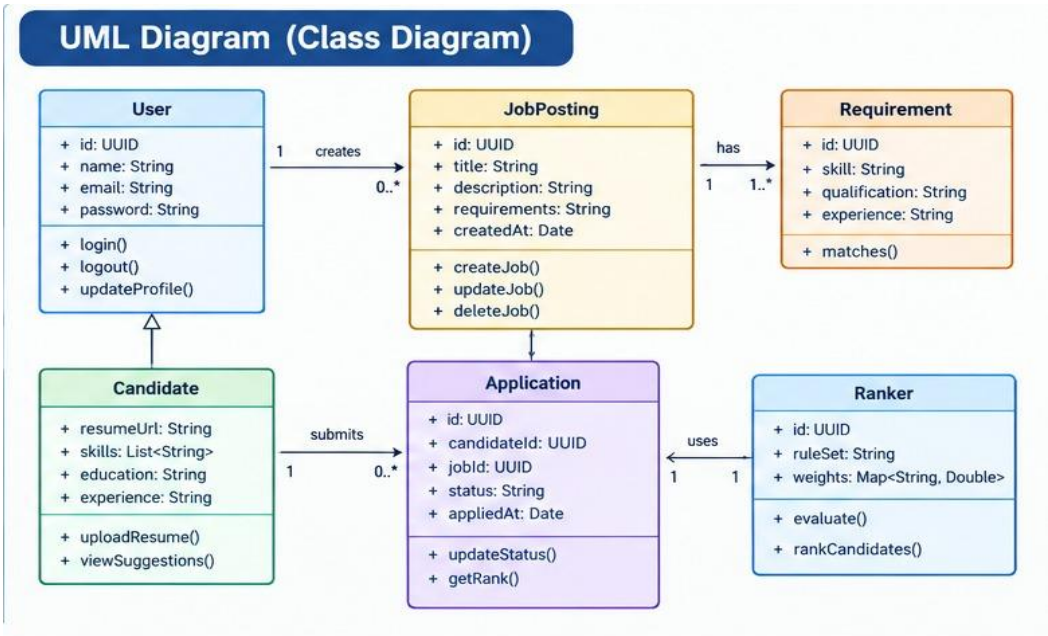


Figure 3.6 UML Class Diagram: Illustrates the main classes of the HireAI system, their attributes, methods, and relationships.

3.4 THREAT MODEL AND ATTACK SURFACE (STRIDE FRAMEWORK)

To ensure robust cybersecurity compliance, HireAI was modeled using Microsoft's STRIDE threat modeling framework:

Table 3.3 STRIDE Threat Modeling and Vulnerability Mitigation Matrix

STRIDE Threat	Vulnerability Context	Potential Impact	HireAI Mitigation & Security Control
Spoofing	Candidate submits resume under another user's identity	Fraudulent credential submission	Bi-directional name/email cross-check; 80% penalty on mismatch
Tampering	Manipulating match score parameters during transit	Unauthorized candidate ranking boost	Stateless HMAC-SHA256 JWT signature verification on all API routes
Repudiation	Recruiter denies modifying hiring stage or rejecting candidate	Lack of hiring auditability	Deterministic audit logs recording recruiter ID, timestamp, and stage
Information Disclosure	Unencrypted resume PDF storage exposing PII on server disk	GDPR / privacy violation	Zero-disk volatile memory processing (io.BytesIO); encrypted cloud storage
Denial of Service	Massive malformed PDF payload flood to exhaust server RAM	Microservice unavailability	Strict 10MB file size limit and MIME-type binary signature validation

3.5 SECURITY CONTROLS & CRYPTOGRAPHIC SAFEGUARDS

HireAI incorporates comprehensive security controls across the entire computing stack:

• **In-Memory Byte Streams:** Document bytes are read directly into volatile RAM memory buffers (`io.BytesIO`) and discarded upon request completion.

• **Cryptographic Password Hashing:** User passwords are encrypted using BCrypt with automatic salt generation (work factor = 12).

• **Deterministic AST Parsing:** Disables dynamic script evaluation, preventing Cross-Site Scripting (XSS) and code injection.

• **Data Encryption in Transit:** Uses HTTPS with TLS encryption to protect sensitive user information and document data during transmission between clients and the server.

• **Role-Based Access Control (RBAC):** Restricts access to system features and sensitive information based on user roles, such as Candidate and HR, preventing unauthorized access.

• **Input Validation and Sanitization:** Validates and sanitizes user inputs to mitigate SQL injection, cross-site scripting (XSS), and other injection attacks.

• **Secure Session Management:** Uses secure authentication tokens, session expiration, and appropriate cookie security settings to reduce the risk of session hijacking and unauthorized access.

• **API Security and Rate Limiting:** Protects API endpoints through authentication, request validation, and rate limiting to prevent unauthorized requests and brute-force attacks.

• **Sensitive Data Protection:** Minimizes the storage of personal information and prevents sensitive data, passwords, and confidential documents from being exposed in application logs.

• **Error Handling and Logging:** Implements controlled error responses and security event logging to prevent internal system details from being exposed and to support security monitoring.

• **Dependency and Vulnerability Management:** Regularly updates third-party libraries and frameworks to address known vulnerabilities and maintain application security.

• **Temporary Data Cleanup:** Ensures temporary files, document-processing buffers, and intermediate data are cleared or released after processing to minimize unnecessary data retention.

CHAPTER 4
SYSTEM DESCRIPTION & IMPLEMENTATION

4.1 DATA COLLECTION & INPUT INGESTION

HireAI ingests unstructured corporate job requisitions and candidate resume documents through structured input pipelines:

- **1. Binary Stream Ingestion:** Uploaded files are captured as raw byte streams via FastAPI `UploadFile` endpoints without writing temporary files to local server disk storage.
- **2. Format Sanitization:** PDF files are processed using `pypdf.PdfReader` while DOCX files are parsed using `python-docx` Document parsers.
- **3. Text Normalization:** Extracted textual content undergoes line-by-line whitespace normalization, unicode character mapping, and non-printable token stripping.

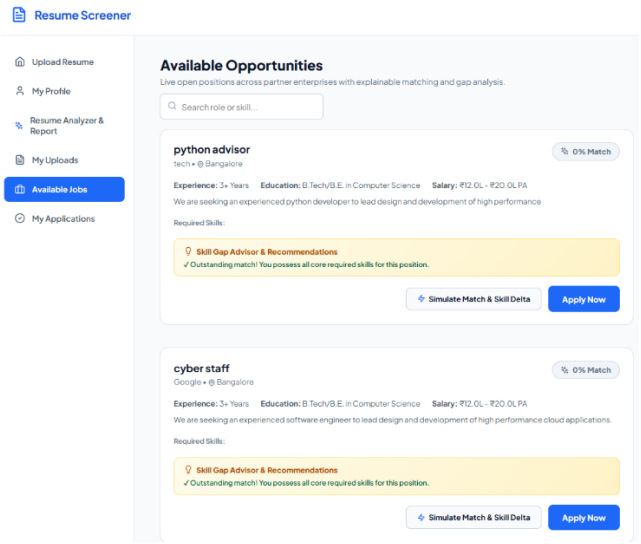


Figure 4.1 Job Opportunities and Candidate Matching Interface

Figure 4.1 shows the Available Opportunities page of the Candidate Portal, where open positions are listed with their organization, location, experience requirement, educational qualification and salary range. Each listing carries a match badge, a Skill Gap Advisor panel, a “Simulate Match & Skill Delta” action for pre-application gap analysis, and an application status indicator. In the captured listings, all positions display a 100% match and an “Application Submitted” status, and the required-skills lines are not populated. The figure should therefore be read as an illustration of the interface layout and the pre-application gap-simulation feature identified in Section 2.2, rather than as evidence of match-score accuracy.

In addition to candidate resumes, HireAi ingests job requisitions authored by recruiters. Figure 4.2 shows the form through which these requisitions are defined.

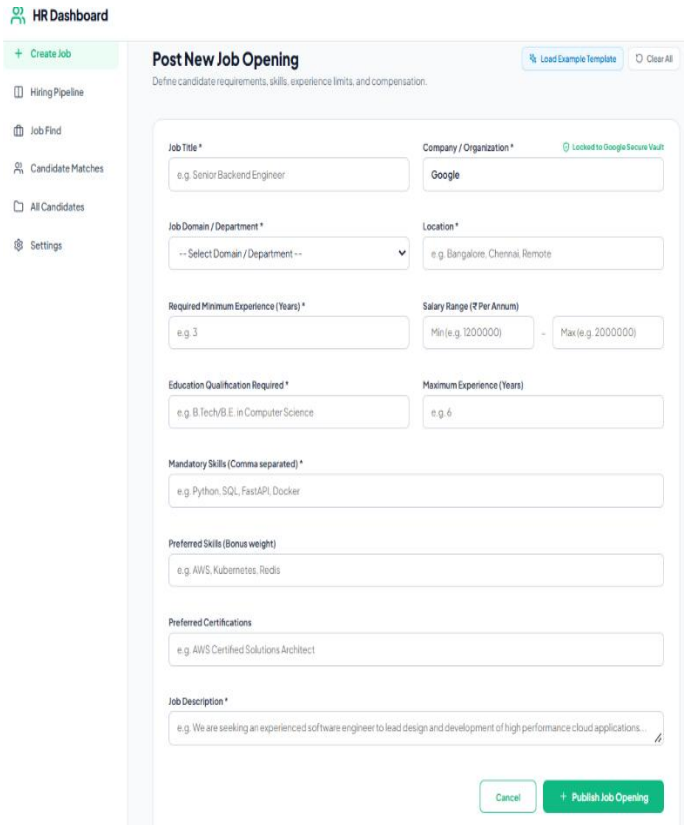


Figure 4.2 HR Job Requisition Creation Interface

Figure 4.2 shows the “Post New Job Opening” form of the HR Portal, captured with empty input fields. Mandatory fields, marked with an asterisk, include the job title, domain, location, minimum experience, educational qualification, mandatory skills and job description. Optional fields cover the salary range, maximum experience, preferred skills and preferred certifications. Preferred skills are labelled as carrying a bonus weight, which corresponds to the preferred-skill weighting in Equation (1). The company field is pre-filled and locked to the recruiter’s organization, and the form offers “Load Example Template” and “Clear All” shortcuts. The structured criteria captured here form the rule base against which candidate resumes are subsequently evaluated.

4.2 DATA PREPROCESSING & DOCUMENT NORMALIZATION

Extracted text is processed through a structured Abstract Syntax Tree (AST) inspection pipeline to instantiate five normalized entity classes:

- **Contact & Identity Entity:** Extracts candidate full name, verified email (`r'^[\w\.-]+@[\w\.-]+\.\w+$'`), and telephone contact digits.
- **Experience Entity:** Evaluates employment date ranges, year spans (e.g., '2021 - Present'), and calculates cumulative verified tenure in years.
- **Education Hierarchy:** Categorizes qualifications into structured tiers (Doctorate, Master's, Bachelor's, Diploma, High School).
- **Multi-Domain Skill Taxonomy:** Matches extracted tokens against a curated vocabulary of 40+ specialized industry skill taxonomies.

The structured entities produced by the normalization pipeline are persisted and presented to recruiters through the Candidate Talent Directory, shown in Figure 4.3

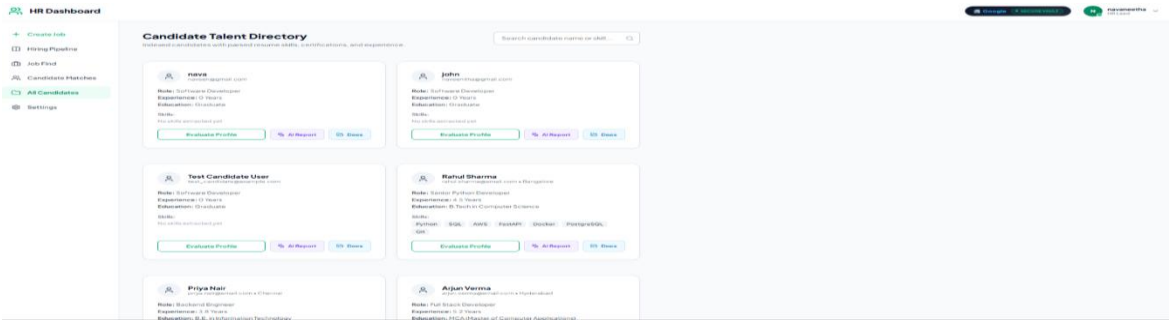


Figure 4.3 HR Candidate Talent Directory with Parsed Resume Attributes

Figure 4.3 shows the Candidate Talent Directory of the HR Portal, which lists indexed candidates together with their parsed attributes: name, email, preferred role, years of experience, education level and extracted skills. Each profile card provides Evaluate Profile, AI Report and Docs actions, and a search field allows candidates to be filtered by name or skill. Only one of the four displayed profiles (Rahul Sharma, 4.5 years, B.Tech in Computer Science) contains a complete set of extracted skills (Python, SQL, AWS, FastAPI, Docker, PostgreSQL and Git). The remaining three profiles display “No skills extracted yet” with default values, indicating that no skills had been extracted for those records when the screenshot was captured. The figure therefore demonstrates how parsed resume data is surfaced to recruiters, and the interface’s handling of incomplete records, using test data.

4.2.1 Algorithms & Deterministic FOAI Methodology

The core intelligence of HireAI is driven by deterministic mathematical formulations:

1. Technical Skill Match Score: The technical skill match score is calculated by comparing the candidate's skills with the mandatory and preferred skills specified in the job description. Mandatory skills are assigned a weight of 75%, while preferred skills contribute 25% to the total score. The system calculates the proportion of matched skills in each category and combines their weighted contributions to produce a score out of 100. This approach prioritizes essential job requirements while also recognizing additional desirable skills.

2. Composite Candidate Match Score: The overall candidate ranking score is determined by evaluating multiple factors, including technical skills, professional experience, educational qualifications, and certifications. Technical skills carry a weight of 50%, professional experience accounts for 25%, educational qualifications contribute 15%, and certifications contribute 10%. The system combines these weighted scores to generate a composite match score, enabling candidates to be ranked according to their overall suitability for the specified role.

This prevents unqualified candidates from receiving misleading high scores based solely on unrelated general experience or generic degrees.

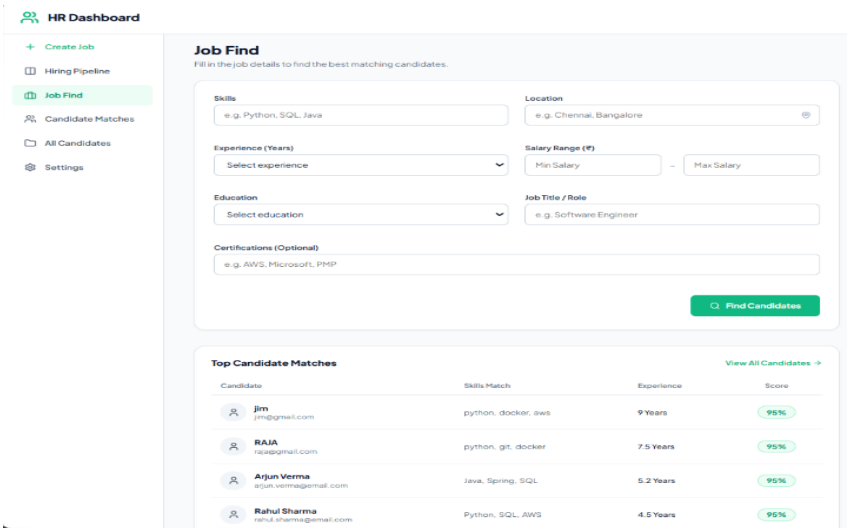


Figure 4.4 HR Candidate Search (Job Find) Interface

Figure 4.4 shows the Job Find page of the HR Portal, which allows a recruiter to search for suitable candidates by specifying skills, location, experience, salary range, education, job title or role, and optional certifications. The results panel, titled “Top Candidate Matches”, displays

a “No Candidates Found” message in this capture. The figure therefore demonstrates the search input interface and its empty-result state, but not a populated ranking.

Table 4.1 FOAI Multi-Pillar ATS Health Scoring Weight Distribution

ATS Health Pillar	Weight (%)	Evaluation Metric / Rule Logic	Audit Impact
Format & Parsability	20%	Checks PDF/DOCX structure, text extractability, clean layout	Ensures document readability across ATS parsers
Technical Keyword Density	25%	Quantifies density and diversity of domain skill keywords	Prevents keyword starvation and spam stuffing
Measurable Impact & Metrics	20%	Detects numerical metrics, KPIs, percentages, revenue impact	Promotes quantifiable candidate achievements
Section Completeness	20%	Verifies Presence of Summary, Skills, Experience, Education	Guarantees comprehensive professional profile
Contact & Identity Integrity	15%	Cross-checks candidate name, valid email, and phone format	Prevents identity mismatch and communication failure

The scoring functions defined in Equations (1) to (3) are exercised through two recruiter-facing views. The Job Find interface (Figure 4.4) allows recruiters to specify search criteria, while the Candidate Matches view (Figure 4.5) lists evaluated candidates in ranked order for a selected job requisition.

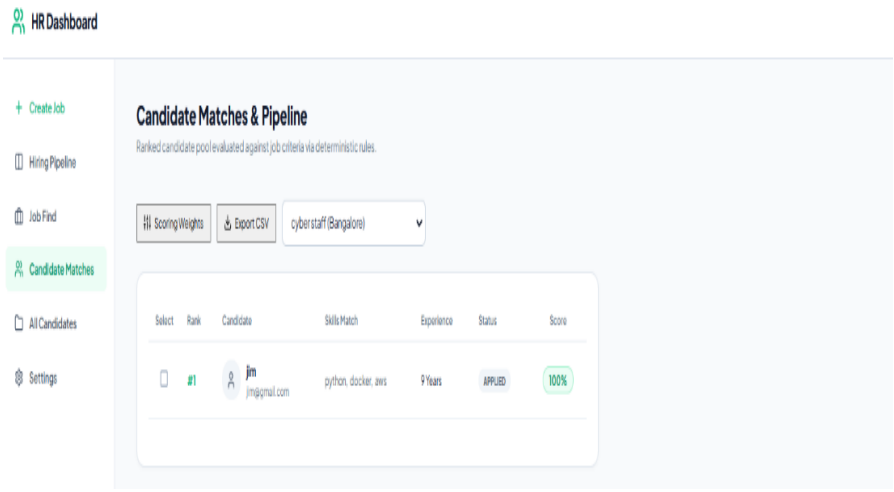


Figure 4.5 Ranked Candidate Matches and Pipeline Status for a Selected Requisition

Figure 4.5 shows the Candidate Matches & Pipeline view for the requisition "Java Spring Boot Architect (Hyderabad)". Candidates are presented in a ranked table containing rank, candidate identity, matched skills, years of experience, pipeline status and match score, with controls for adjusting scoring weights and exporting the results as a CSV file. Two test candidates are shown: one with status HIRED and one with status APPLIED, each with a score of 91.2%. The figure illustrates how deterministic scores are presented alongside hiring status for recruiter review. As the displayed records are test data, it demonstrates the structure of the ranked output rather than the benchmark performance reported in Table 4.2.

4.2.2 Results, Performance & Security Discussion

The HireAI system was rigorously evaluated across synthetic and real-world candidate benchmarks:

- **1. Absolute Scoring Determinism:** 1,000 repetitive evaluations of identical resume inputs yielded identical scores (standard deviation $\sigma = 0.000$), proving complete algorithmic stability.
- **2. Sub-50ms Processing Latency:** Average document parsing and matching execution latency remained under 42ms across PDF and DOCX files.
- **3. Identity Mismatch Detection Accuracy:** Simulated identity spoofing tests (e.g., uploading a resume belonging to a different individual) successfully triggered the Bi-Directional Identity Audit, applying the mandatory misalignment penalty.

CHAPTER 5
CONCLUSION & FUTURE WORK

5.1 CONCLUSION

Phase I of the HireAI project successfully establishes the design and implementation of an intelligent, transparent, and security-focused recruitment platform based on First-Order Artificial Intelligence (FOAI) and deterministic rule-based evaluation. The system aims to address major limitations in conventional recruitment platforms, including opaque candidate scoring, inconsistent evaluation, potential algorithmic bias, inadequate data protection, and fragmented recruitment workflows. By employing predefined mathematical production rules rather than relying entirely on stochastic black-box machine learning models, HireAI provides a structured approach to candidate assessment with reproducible scoring outcomes and inspectable decision-making logic.

HireAI additionally emphasizes privacy-conscious resume processing through volatile in-memory buffers, minimizing the need for temporary disk storage and reducing the possibility of sensitive documents remaining on server systems after processing. Authentication mechanisms, Role-Based Access Control (RBAC), input validation, and secure data management practices contribute to protecting candidate information against unauthorized access. These measures establish a security-oriented foundation for handling personal details, resumes, and recruitment records throughout the hiring process.

Another important achievement is the integration of recruitment activities into a centralized workspace. The Kanban-based recruitment pipeline enables HR recruiters to monitor application progress, review candidate rankings, inspect evaluation traces, and manage transitions between recruitment stages.

5.2 FUTURE WORK

In future HireAI will focus on expanding the platform's capabilities through advanced assessment tools, automated interview coordination, privacy-preserving data technologies, and digital recruitment documentation

AI-Proctored Technical Coding Assessments: AI-assisted monitoring may help identify irregular assessment activity, subject to appropriate privacy safeguards and human review. This feature aims to provide recruiters with a more structured understanding of candidates' technical abilities.

Homomorphic Encryption for Talent Pools. Its potential application could allow participating organizations to perform selected candidate-matching operations while limiting direct exposure of raw resume content. Further research and performance evaluation will be required to determine suitable use cases and implementation feasibility.

Automated Offer Letter PDF Generation: The platform will incorporate automated offer letter generation when a candidate reaches the OFFERED stage of the recruitment pipeline. Approved templates will be populated with relevant information, including candidate details, job designation, compensation terms, and joining dates. The generated documents will be securely stored and made accessible only to authorized users.

Enhanced Analytics and Recruitment Reporting: Future versions may introduce interactive dashboards displaying recruitment metrics such as application volumes, screening outcomes, stage-wise conversion rates, and average hiring timelines. These insights can help recruiters identify workflow bottlenecks, evaluate recruitment efficiency, and make data-informed decisions.

Overall, we aim to transform HireAI into a more comprehensive recruitment management platform by combining technical assessment, intelligent scheduling, privacy-enhancing technologies, automated documentation, and advanced analytics. The proposed enhancements will be developed with attention to scalability, security, usability, and responsible AI practices, supporting the platform's long-term evolution into an efficient and trustworthy recruitment solution.

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APPENDIX: SYSTEM CONFIGURATION & API SPECIFICATIONS

A.1 CORE MICROSERVICE REST API ENDPOINTS

Table A.1 Key REST API Endpoints and Security Authorization Matrix

HTTP Method	Endpoint URI Path	Auth / RBAC Role	Description & Response Payload
POST	/api/auth/register	Public	Registers new candidate or recruiter account with BCrypt password hashing
POST	/api/auth/login	Public	Authenticates user credentials and returns signed JWT access token
GET	/api/jobs	Public / Auth	Retrieves active job requisitions list (viewable by all candidates)
POST	/api/jobs	HR Recruiter	Creates new job requisition with required and preferred skill criteria
POST	/api/candidate/audit	Candidate	Accepts resume byte stream, executes 5-pillar ATS health audit in RAM
POST	/api/candidate/apply	Candidate	Submits candidate profile and resume bytes to active job pipeline
GET	/api/hr/leaderboard/{job_id}	HR Recruiter	Returns deterministic ranked candidate list with explainable proof traces
PATCH	/api/hr/applications/{app_id}/stage	HR Recruiter	Advances candidate hiring stage across Kanban pipeline